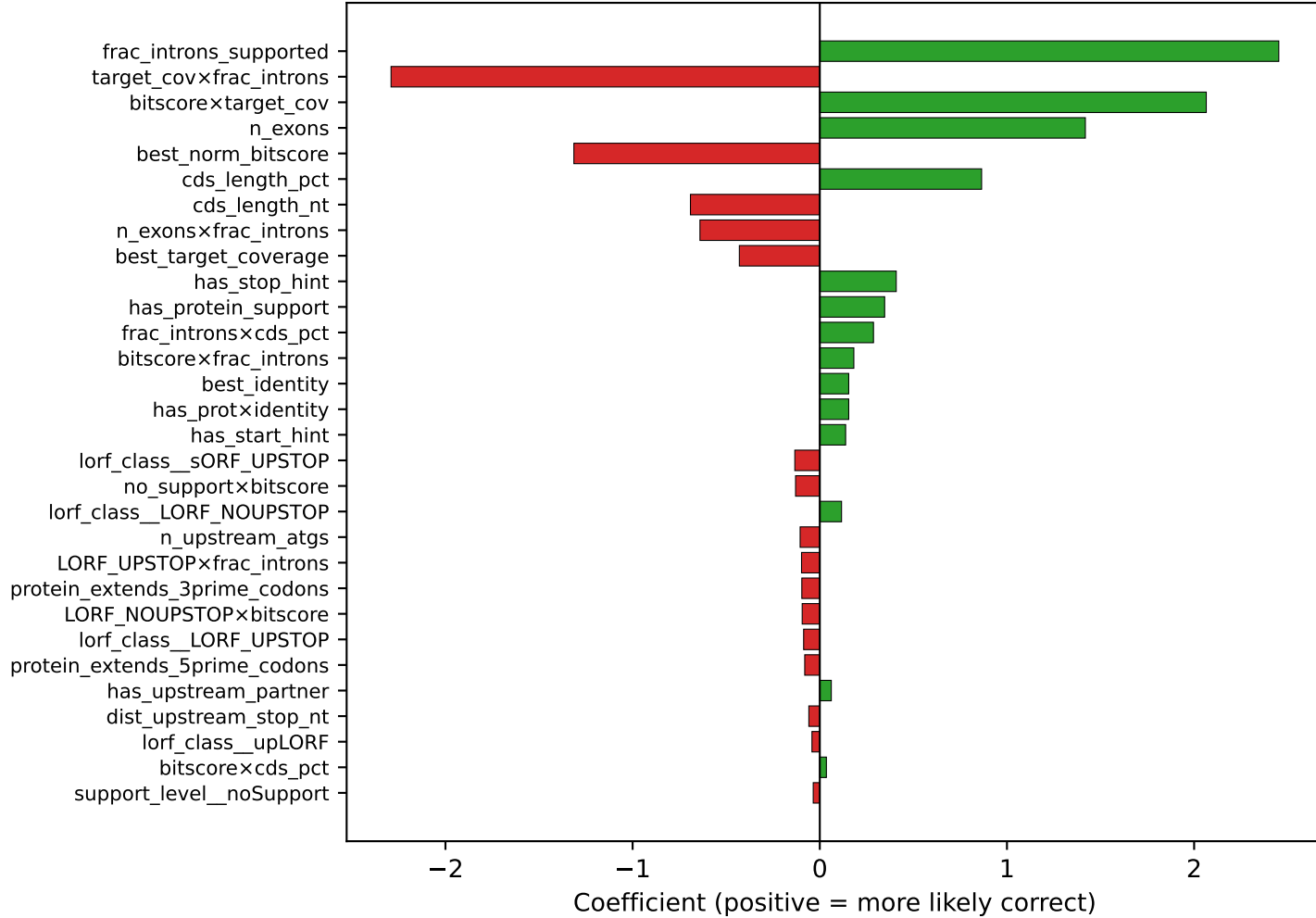
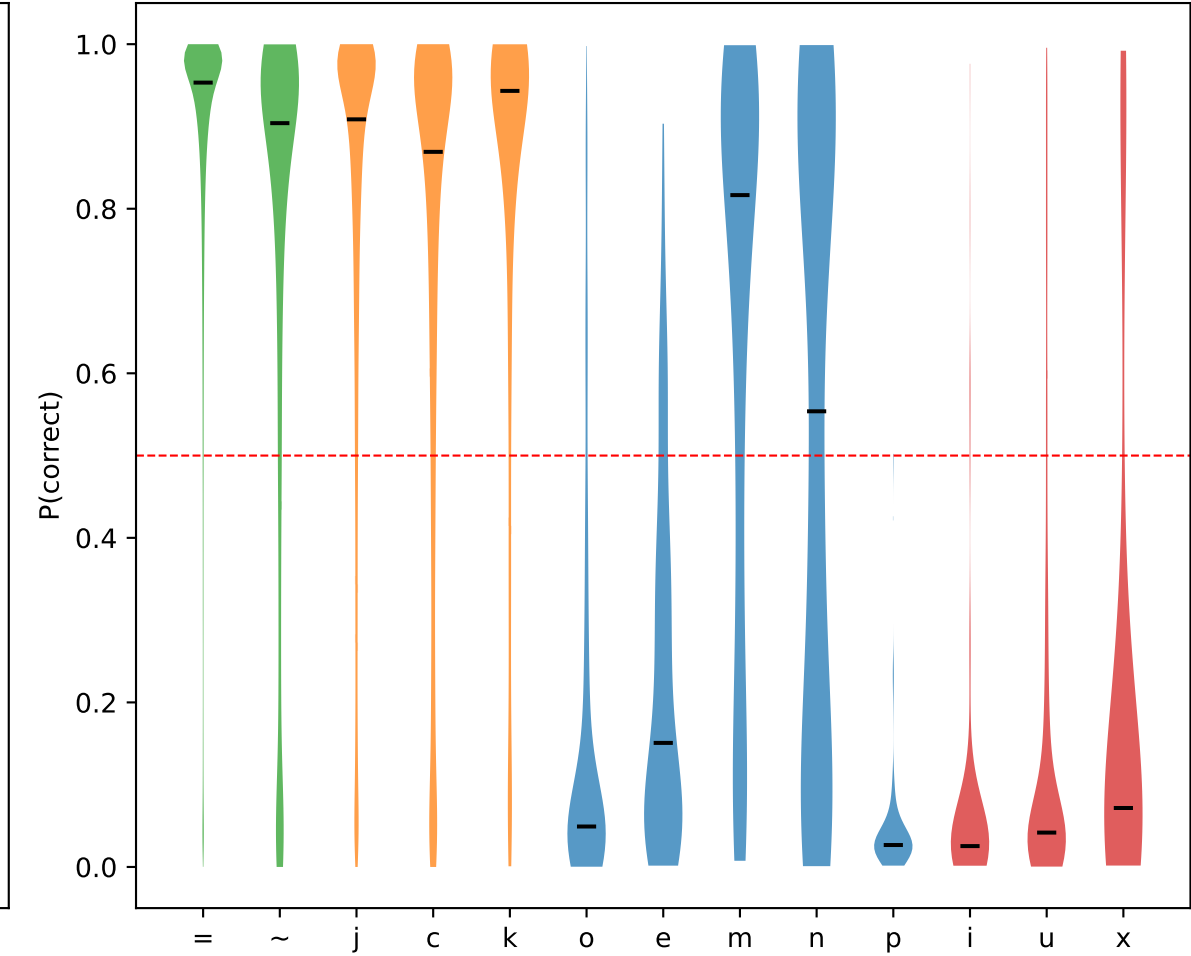


Logistic Regression (AUC=0.9577)

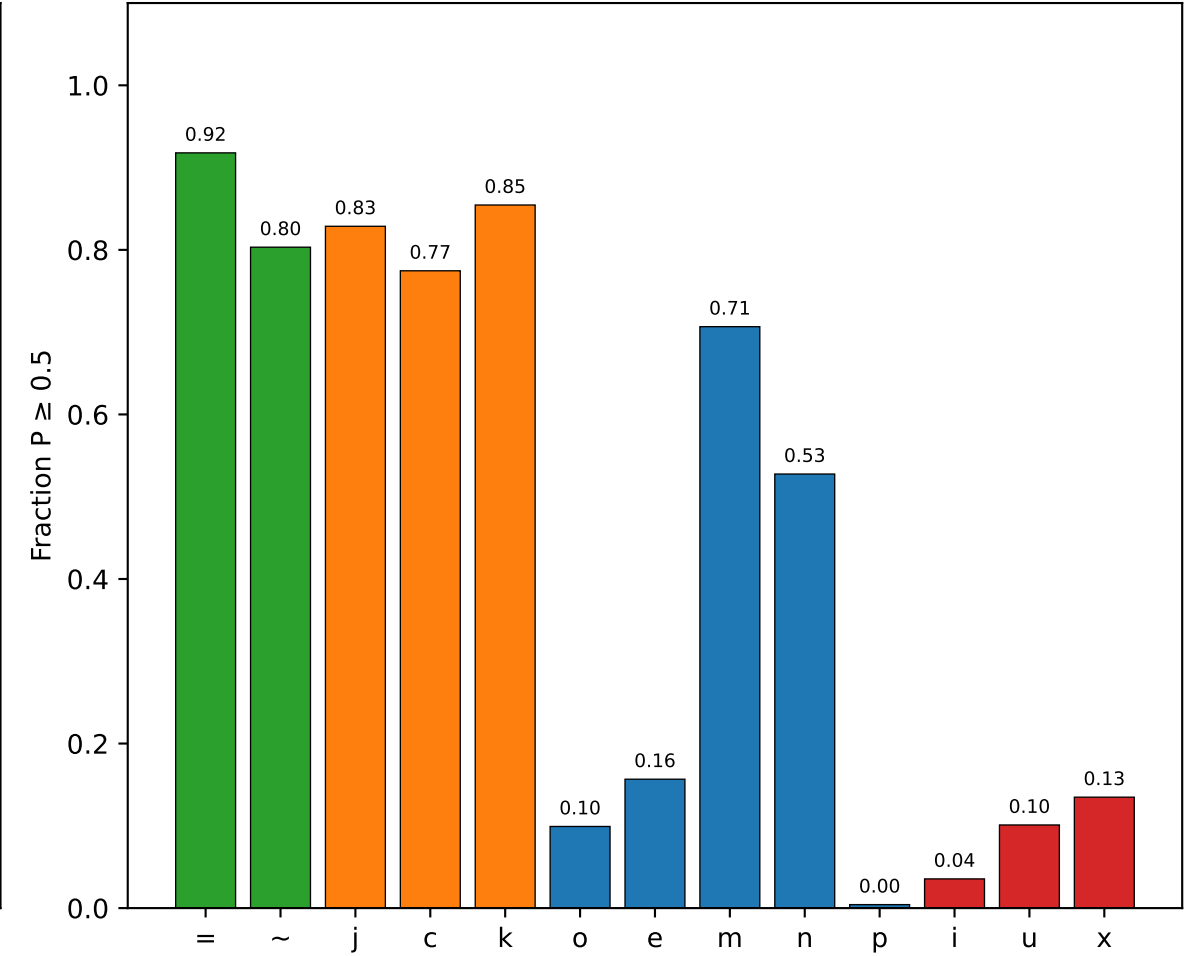
Top-30 coefficients (green=correct, red=wrong)



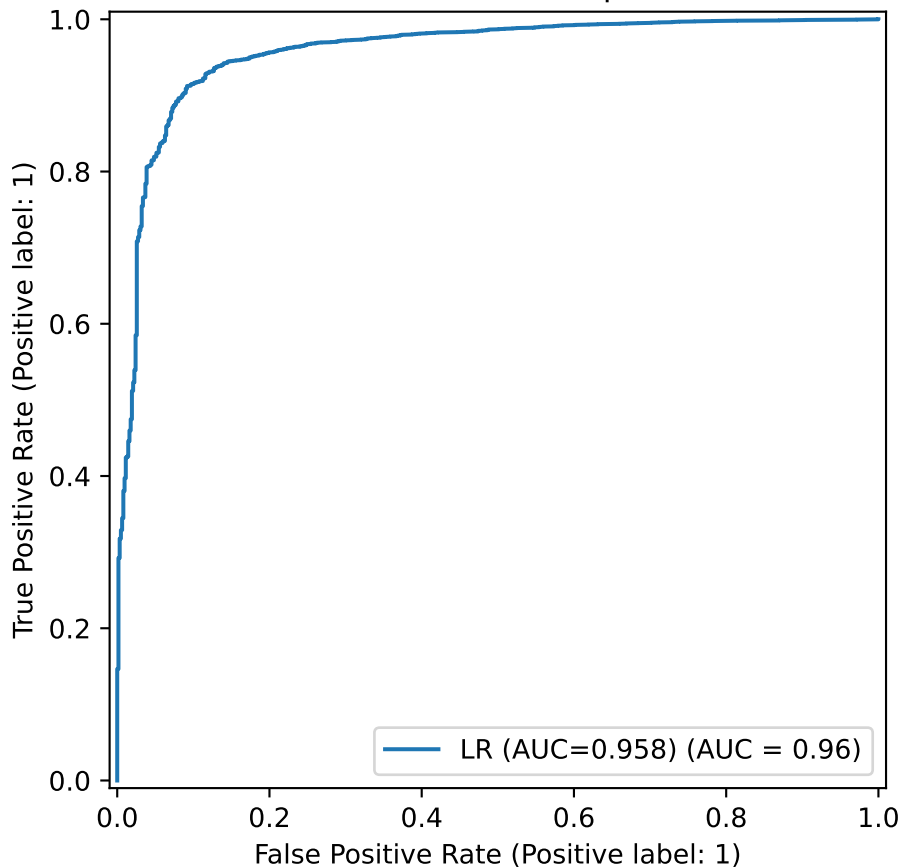
Score distribution per class



Fraction rescued at P≥0.5

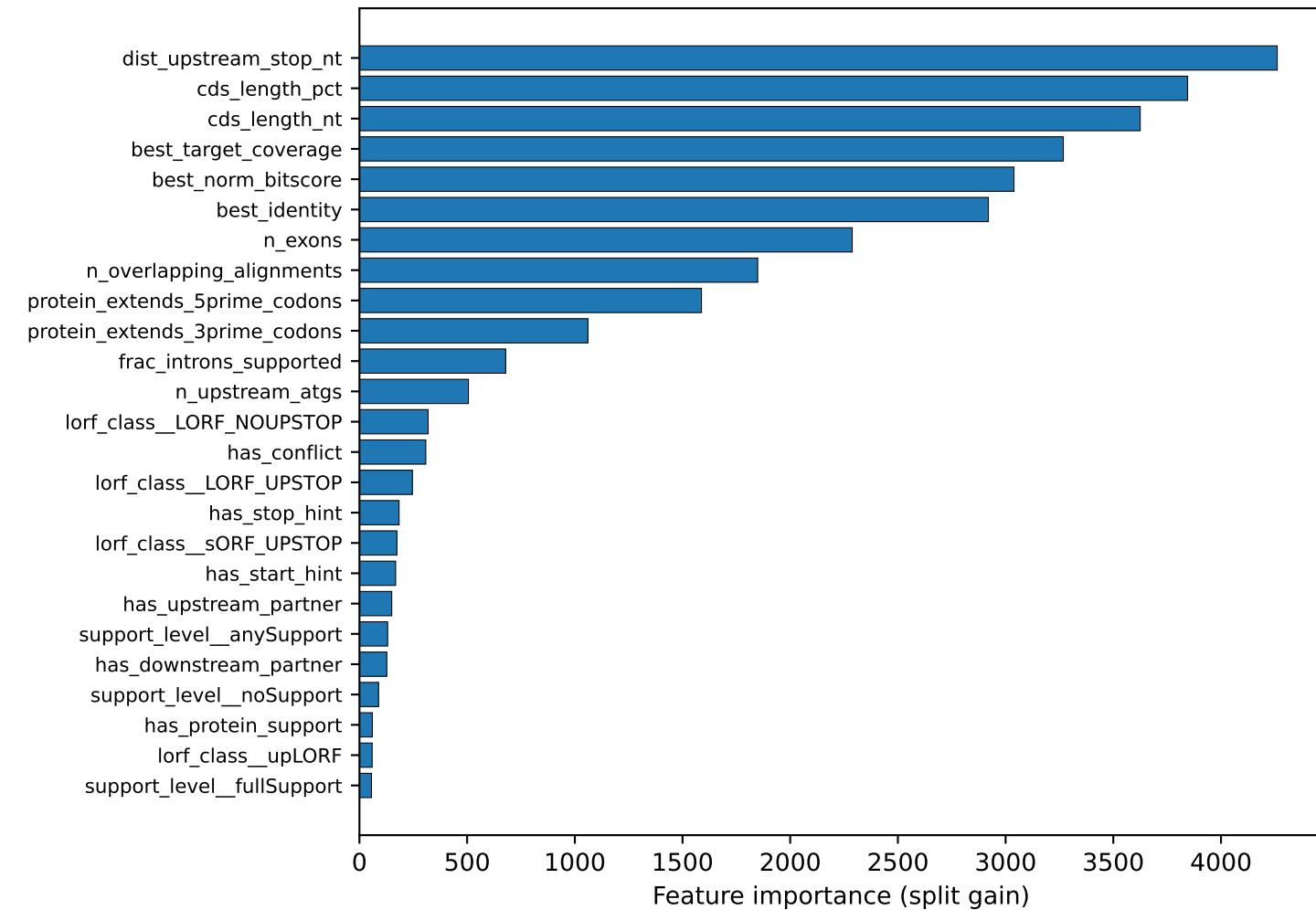


LR ROC — test split

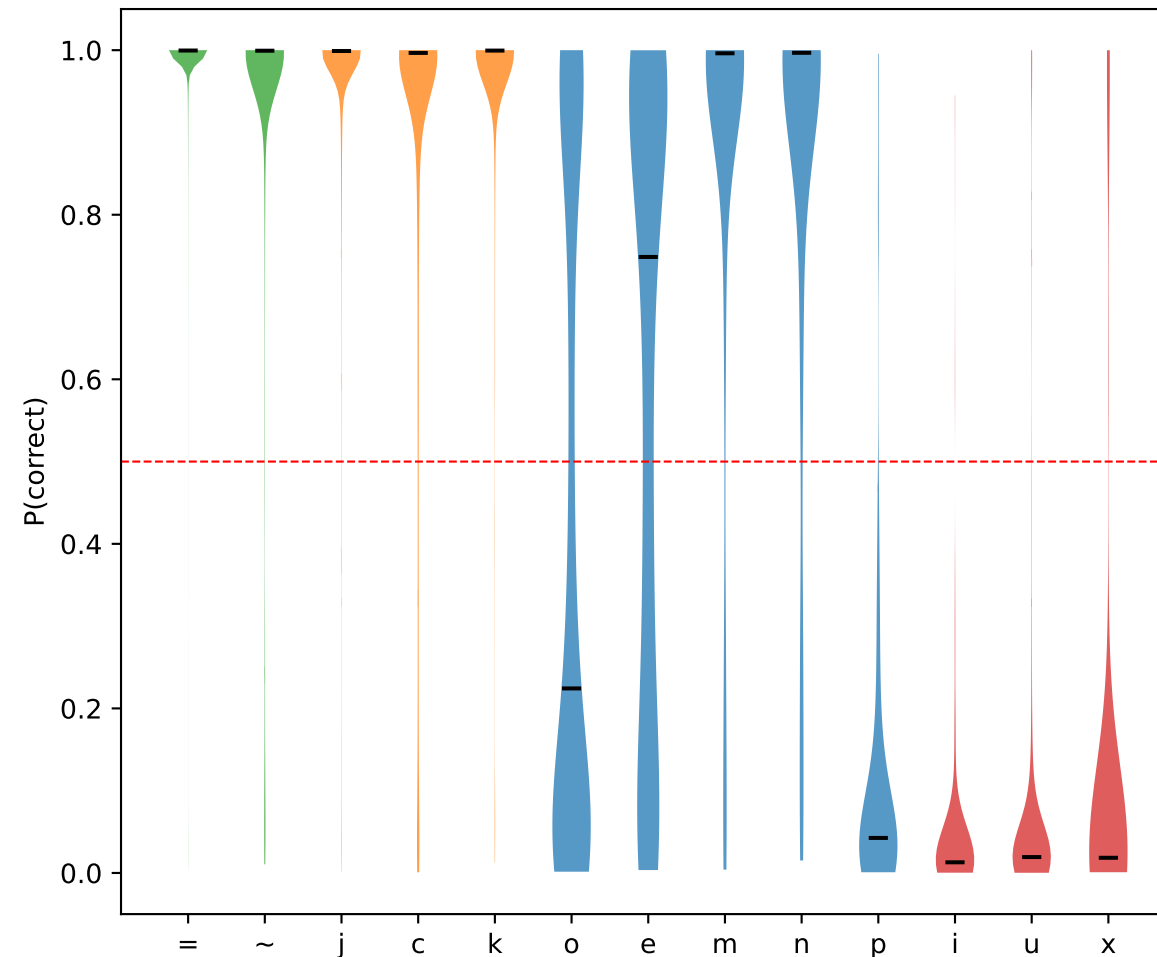


LightGBM (AUC=0.9658)

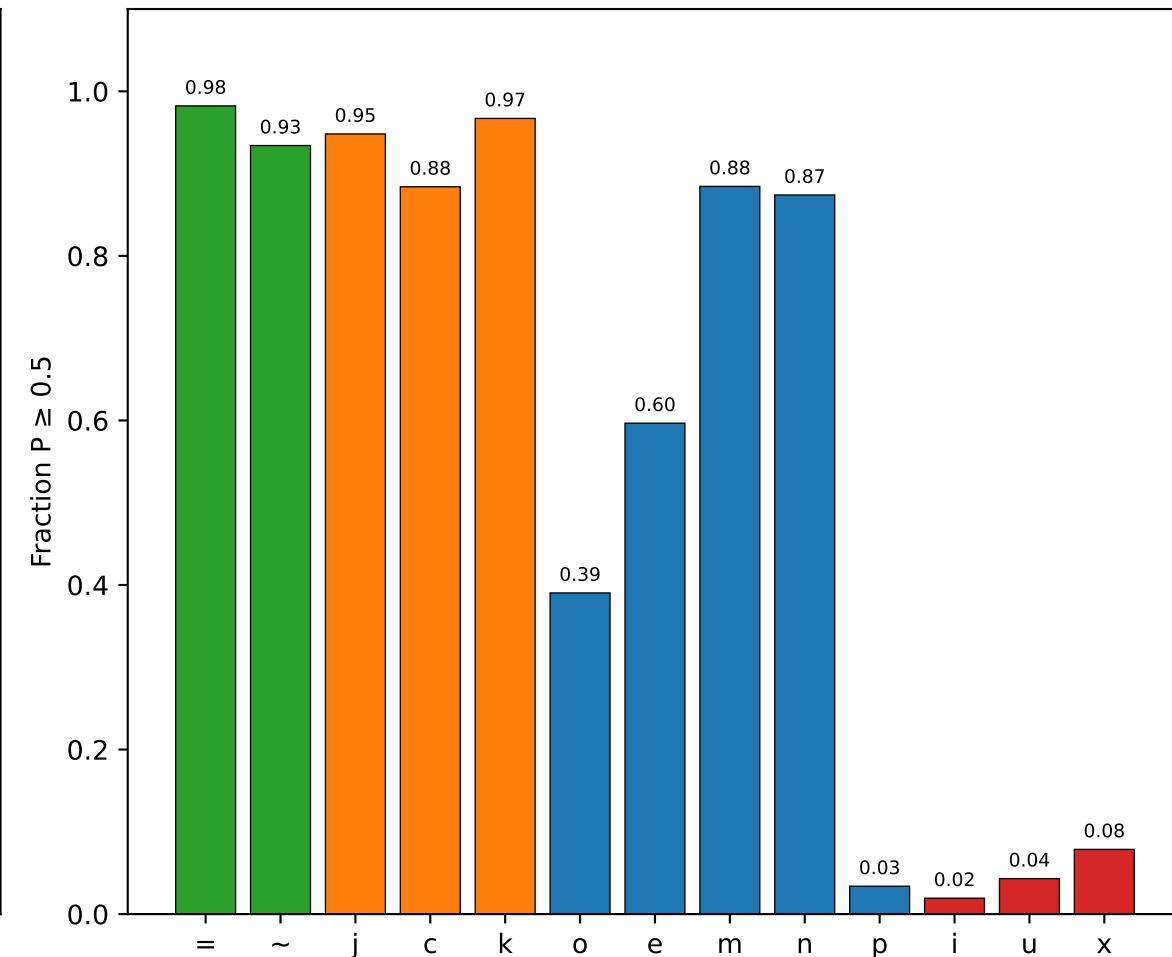
Top-25 features



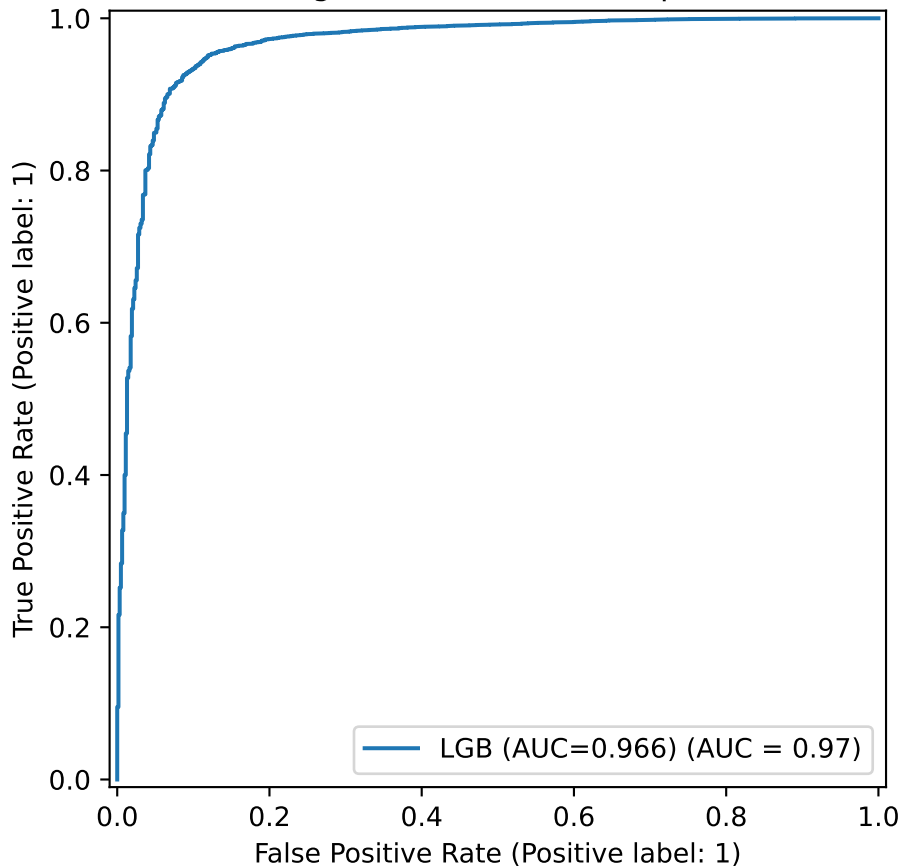
Score distribution per class



Fraction rescued at P≥0.5



LightGBM ROC — test split

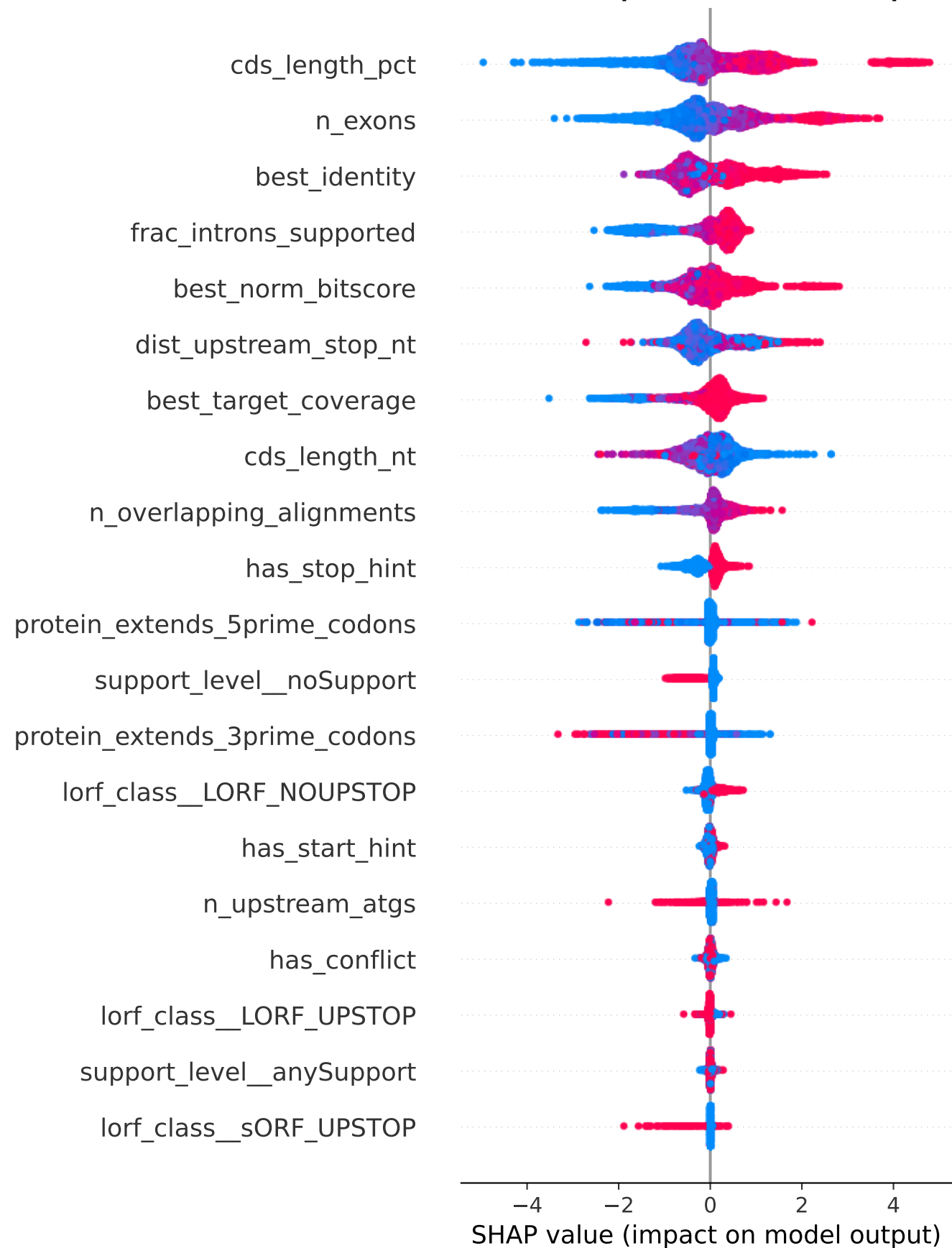


SHAP beeswarm — positive class (correct prediction)

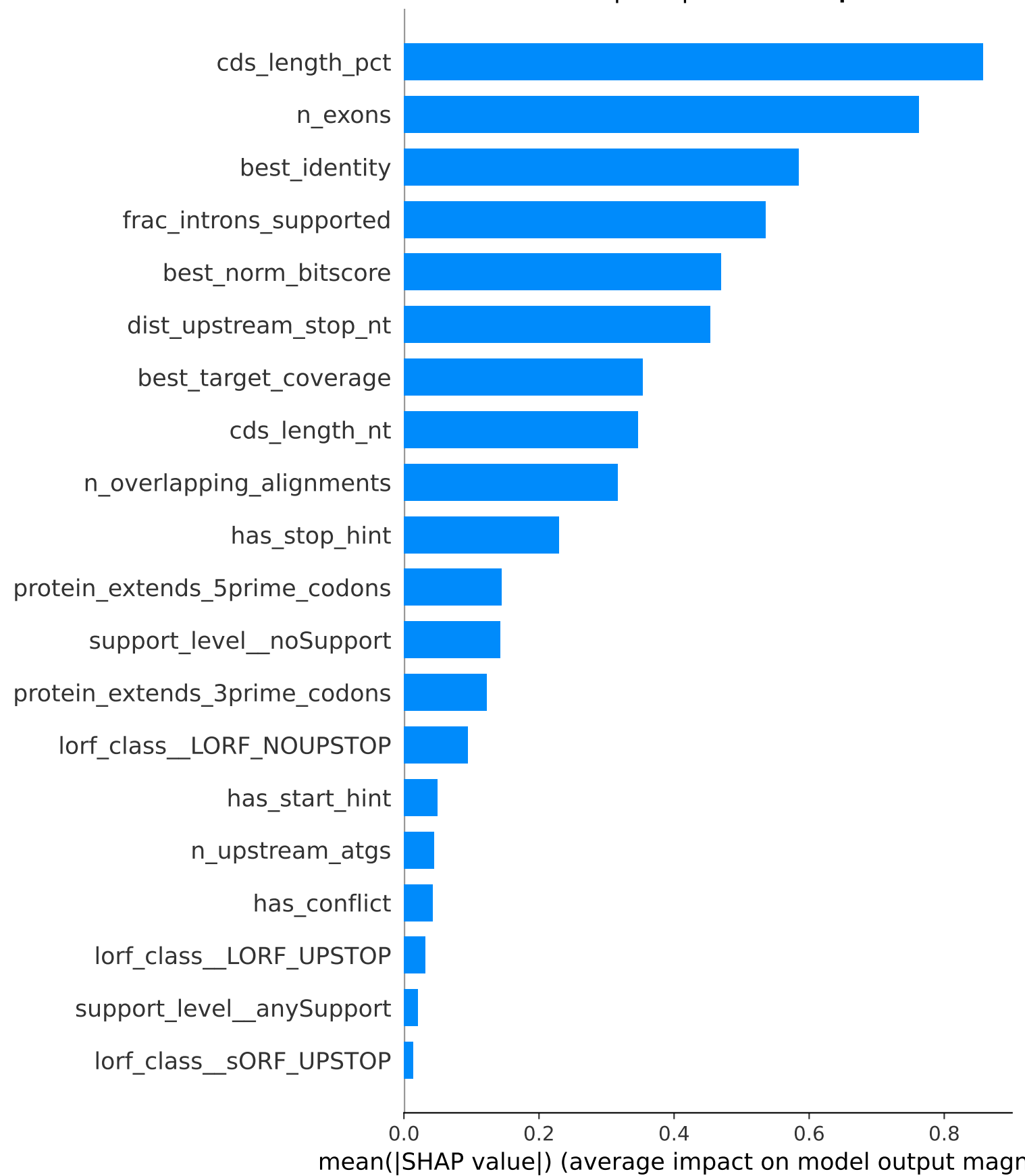
High

Feature value

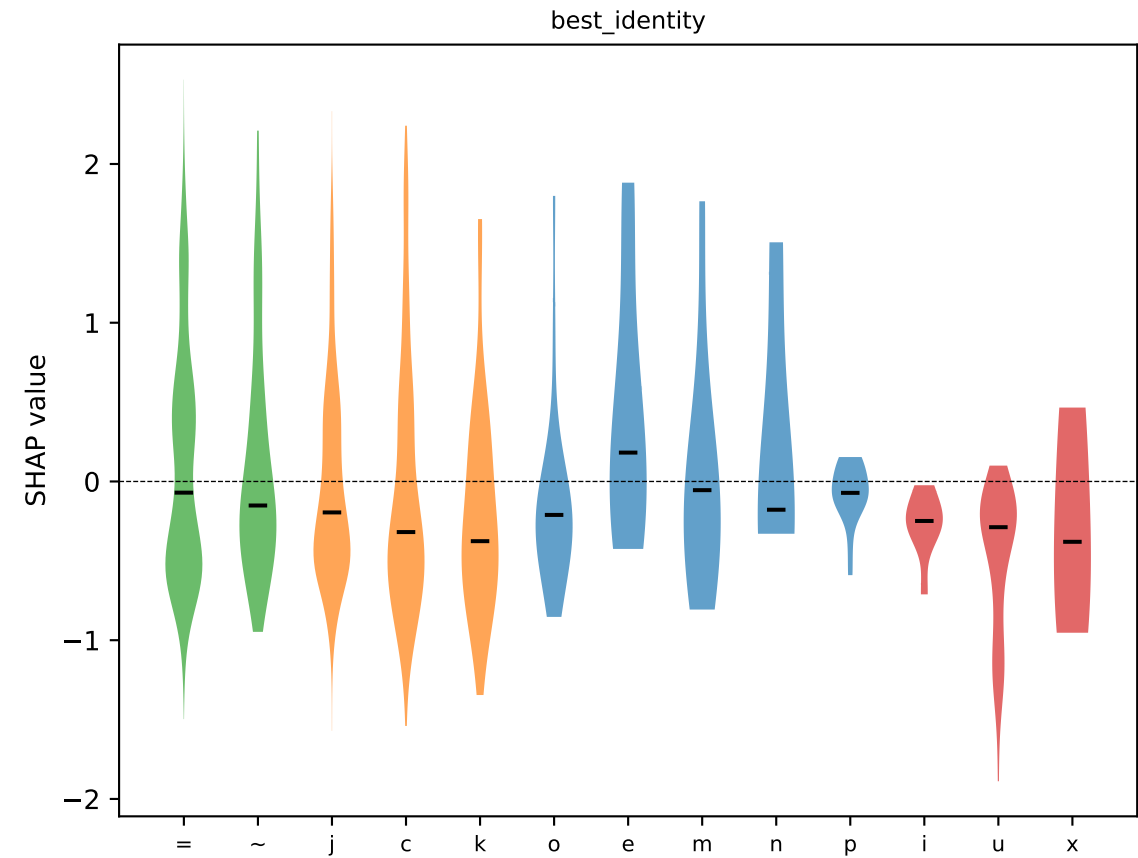
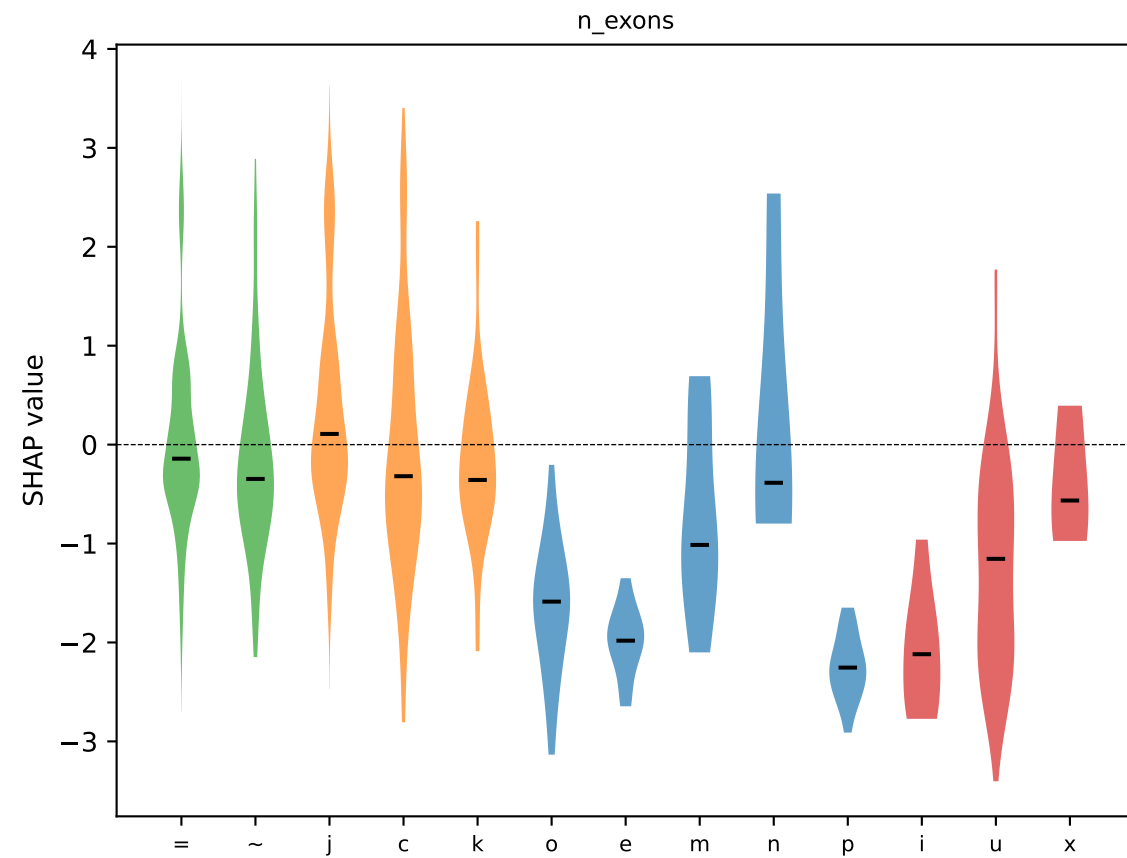
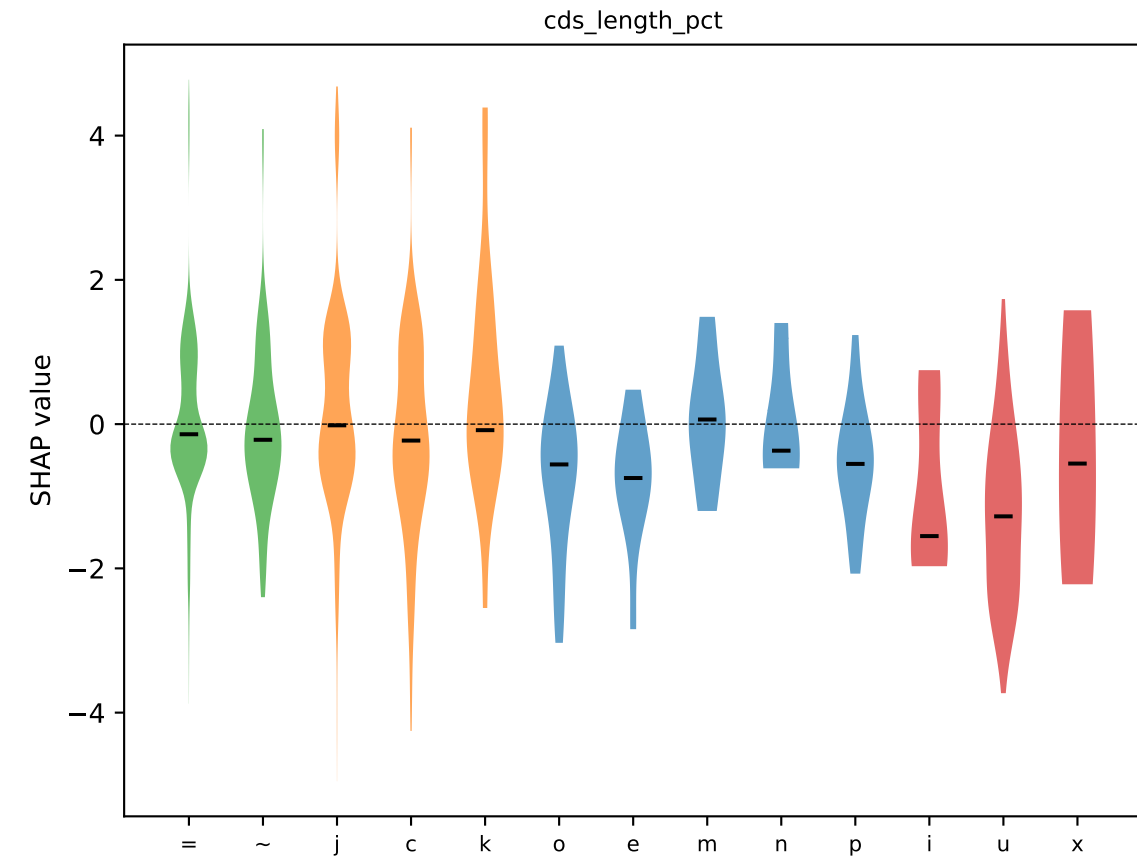
Low



SHAP mean |value| — feature importance



SHAP value distributions per gffcompare class (top-3 features)



Model comparison: Logistic Regression vs LightGBM

